

Practice Session 01 – Report

Cytoscape: editing and visualizing networks (Due Sept. 29th, 18:30)

1. Description of the three networks

- 1.1. Zachary's Karate Club network: describes 34 people that belong to a karate club (nodes) and the relationship between pairs that interacted outside the club (edges). The interesting thing about this network is that, because of a conflict between nodes 34 and 1 (president and sensei respectively) the club ended up splitting in two almost separate groups that barely interacted between them.
- 1.2. Star Wars Characters network: captures 110 nodes that represent characters in the Star Wars movies that are linked through an edge if they are mentioned, or they appear in the same scene simultaneously. There are a total of 444 links in this network.
- 1.3. Network of Tortoises network: contains 787 nodes representing tortoises, which are linked if they have had some kind of social interaction between any pair of them. The number of social interactions in this set of tortoises is 1713.

Network	Nodes	Edges
Karate Club	34	78
Star Wars	110	444
Tortoises network	787	1713

Table 1: Nodes and edges of the three networks

2. Zachary's Karate Club

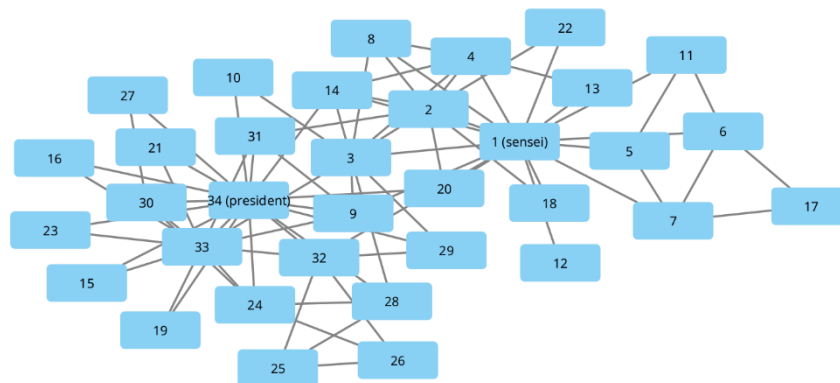


Figure 1: Zachary's Karate Club visualized.

The top three nodes having the larger degree are: node 34 (president) with Degree=17, node 1 (sensei) with Degree=16, and node 33 with Degree=12.

We can visualize the network this way thanks to the use of the Compound Spring Embedder layout, which is an algorithm that makes visualizing the network look aesthetically good. A thing about this algorithm that differentiates it from the force-directed graph layout algorithms is that it can handle compound nodes (nested nodes).

The three nodes with the largest closeness centrality are: node 1 (sensei), node 3, and node 34 (president) in descending order.

The three nodes with the largest betweenness centrality are: node 1 (sensei), node 34 (president), and node 33 in descending order.

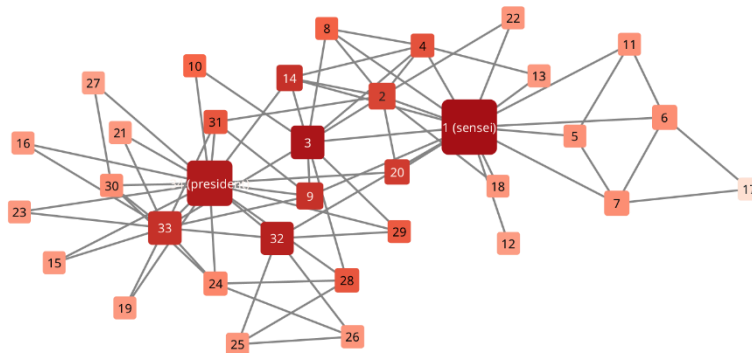


Figure 2: Mapping closeness centrality and betweenness centrality to color and sizes proportionally.

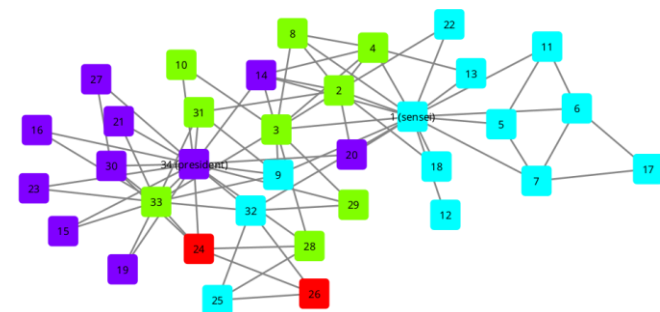
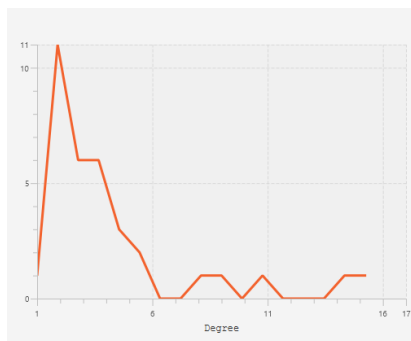


Figure 3: Degree distribution of Zachary's Karate Club. Figure 4: Discrete mapping using as a clustering method Fast Greedy.

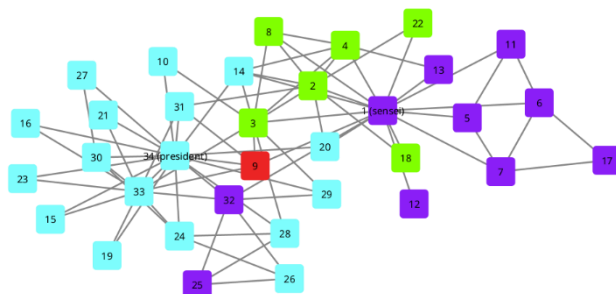


Figure 5: Discrete mapping using as a clustering method the Leading Eigenvector method.

In both clusterings we can see that there are two very distinct groups on the left and the right of the network. Especially in the Leading Eigenvector clustering we can see that there is a left cluster (light blue) and a right cluster (purple) that have almost no connections between them. There are some nodes in between that act as “intermediates” between those.

3. Star Wars Characters

There are a couple of characters that have degree ≥ 25 . For instance, Padme (as annotated in the network). This means that there are more than 25 mentions or appearances of other characters simultaneously in one or more scenes of the Star Wars movies. From that we can interpret that those characters are the main ones. In Figure 8 we have applied Affinity Propagation clustering, but no clue on what it is supposed to represent.

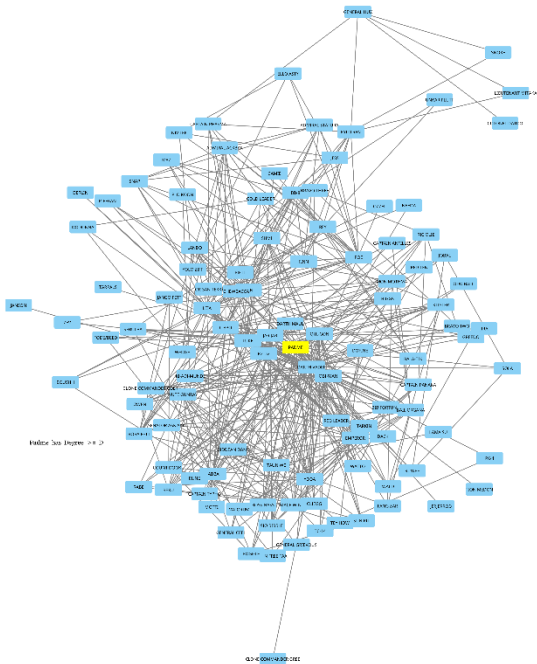


Figure 6: Star Wars Characters network visualized.

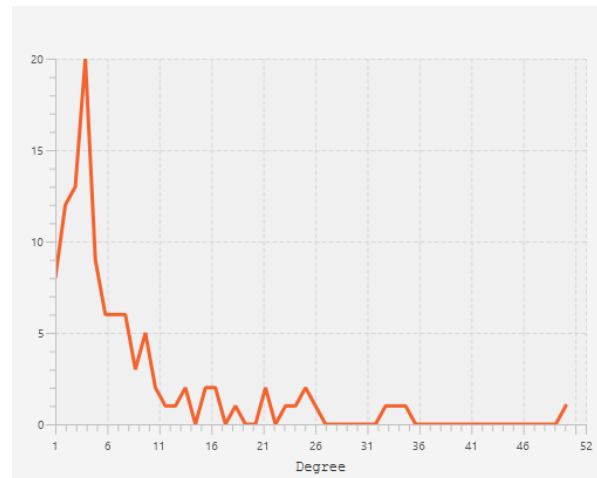


Figure 7: Degree distribution of Star Wars Characters network.

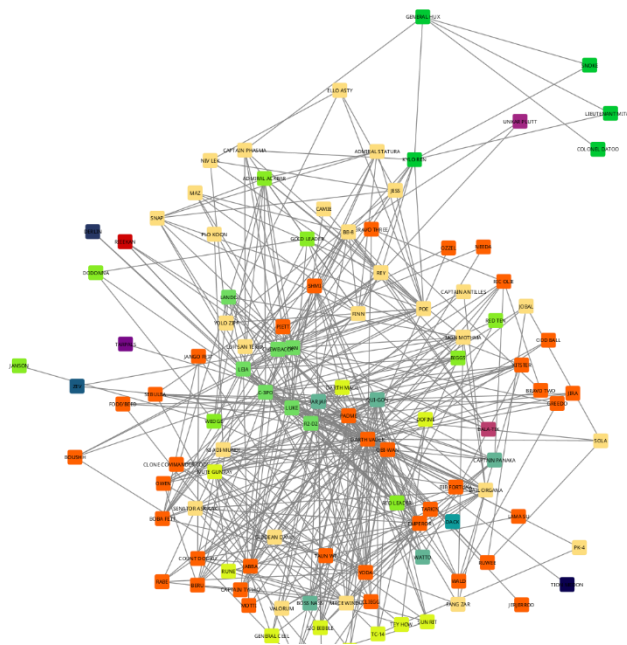


Figure 8: Discrete mapping using as a clustering method the Affinity Propagation method.

4. Network of Tortoises

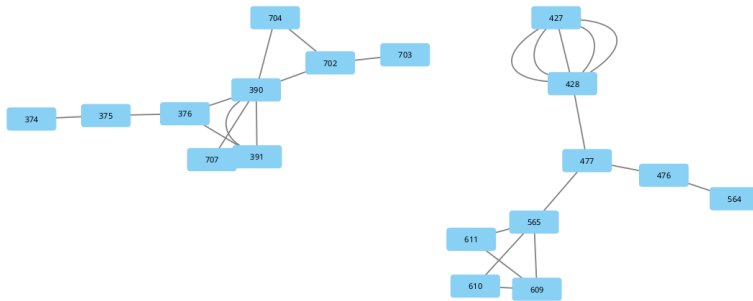


Figure 9: Second and third largest connected components in Tortoises network, both with nine nodes.

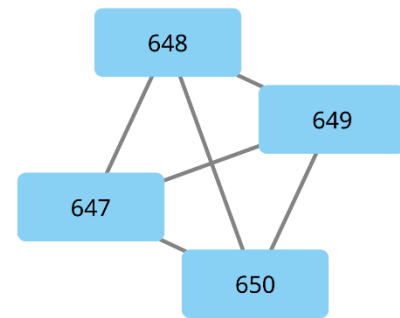


Figure 10: Largest clique (with 4 nodes).

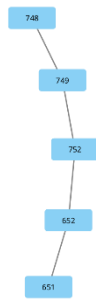


Figure 11: Largest line graph (with 5 nodes).

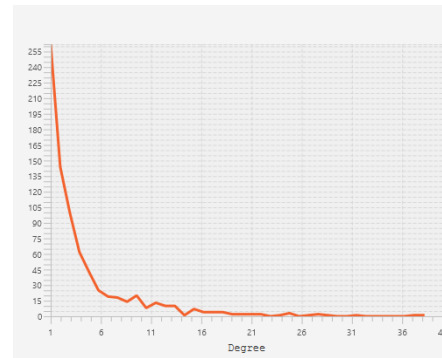


Figure 12: Degree distribution of the Tortoises network

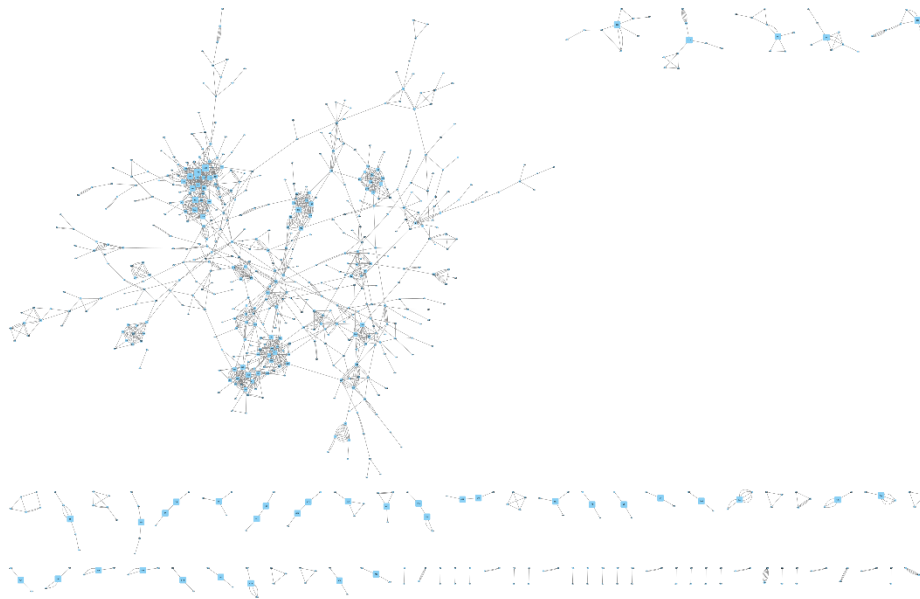


Figure 13: Mapping degree and betweenness centrality to the size of the nodes proportionally

I hereby declare that all of the text, tables, and figures in this report were produced by myself.